Day 11

11 of 30 · Sundays holiday

Blood & platelets — the science behind PRP

బ్లడ్ & ప్లేట్‌లెట్స్ — PRP సైన్స్

TODAY'S OBJECTIVES · ఈ రోజు లక్ష్యాలు

- Explain what PRP is and how it stimulates hair growth
- Understand blood components and what the centrifuge separates
- Prepare the theory you need before touching a tube

THEORY · థియరీ

Blood components

Whole blood = plasma (55%), buffy coat (platelets + white cells) and red cells (45%). Spinning separates them by density. Platelet-rich plasma is plasma with platelets concentrated 3–6× above baseline. Platelets contain alpha granules full of growth factors — PDGF, VEGF, EGF, IGF-1, TGF-β.

How it works on hair

Injected around the follicles, the growth factors increase dermal papilla cell activity and blood supply, prolong anagen, and reduce miniaturisation. Effect: less shedding by 4–6 weeks, visible new growth by 3 months, density improvement by 5–6 months. It supports — it does not replace — the doctor's medical therapy, and it cannot regrow follicles that are already scarred.

Tubes and activation

PRP tubes contain an anticoagulant (ACD-A or sodium citrate) and often a separating gel. Handle gently — rough shaking damages platelets. Some protocols activate with calcium chloride; many hair protocols use non-activated PRP. Always follow the doctor's chosen protocol exactly, including spin speed and time.

Hands-on today

- Identify tube types, anticoagulants and their uses
- Watch a full spin and identify the layers in the tube
- Write the clinic's exact PRP protocol from observation

Safety & contraindications

- Never use a tube past its expiry or with a broken seal
- Label every tube with the client's name immediately after draw — no exceptions

KEY POINTS TO REMEMBER · గుర్తుంచుకోవలసినవి

- Platelets concentrated 3–6×
- Growth factors: PDGF, VEGF, IGF-1
- Shedding reduces by 4–6 weeks
- Follow the clinic's spin protocol exactly

Homework

Draw a labelled tube after centrifugation and write which layer is used.

Grooming & professional tip

One mislabelled tube can mean someone else's blood in a client. Label at the chair, in front of the client, every single time.

SELF-CHECK · సెల్ఫ్ చెక్

- Did I follow hygiene and consent for every client today?
- Which step did I need help with — and what will I drill tonight?
- Did I write my records and settings before leaving?

Trainer sign: _____ Trainee sign: _____

MY NOTES · నా నోట్స్